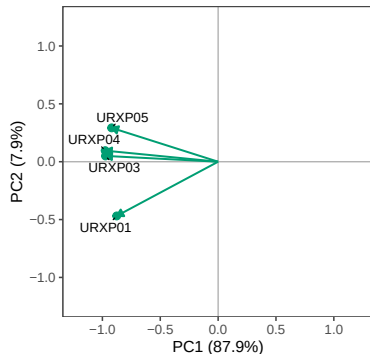


Per-modality PCAs

PC1/PC2 variable loadings, one panel per modality block

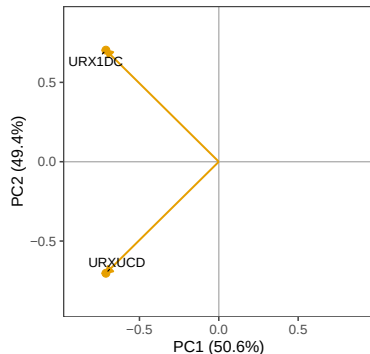
Urinary PAH metabolites (U-PAH)

N = 1,800; vars used = 4; PC1 = 87.9%; PC2 = 7.9%



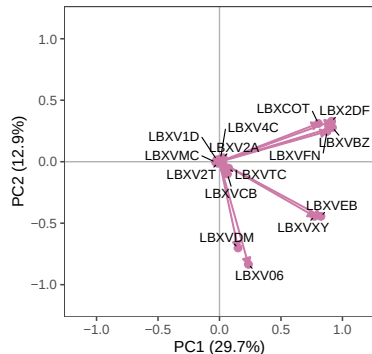
Urinary metals & mercapturates (U-MM)

N = 1,788; vars used = 2; PC1 = 50.6%; PC2 = 49.4%



Blood VOCs & combustion analytes (B-VOC)

N = 1,316; vars used = 15 (dropped 2 zero-variance LOD-pinned vars)



Four independent PCAs, one per modality block.

Q (single var) skipped – PCA on a single column is undefined. Each remaining panel uses the standard prcomp route on the within-modality complete-case subset (NHANES sub-samples DO NOT include all 24 vars).

U-PAH: N = 1,800 participants with all 4 PAH metabolites measured; prcomp on standardized 1,800 × 4 matrix.

U-MM: N = 1,788 with all 2 urinary metals/mercapturates; prcomp on standardized 1,788 × 2 matrix.

B-VOC: N = 1,316 with all 16 blood VOCs / combustion analytes (2 zero-variance LOD-pinned vars dropped from the subset: LBXV3B, LBXV4E); prcomp on standardized 1,316 × 15 matrix.

Each panel plots PC1 vs PC2 variable loadings (eigenvector × sqrt(eigenvalue)). Joint complete-case N across ALL 24 vars = 0 – saturated 24-var PCA is infeasible.